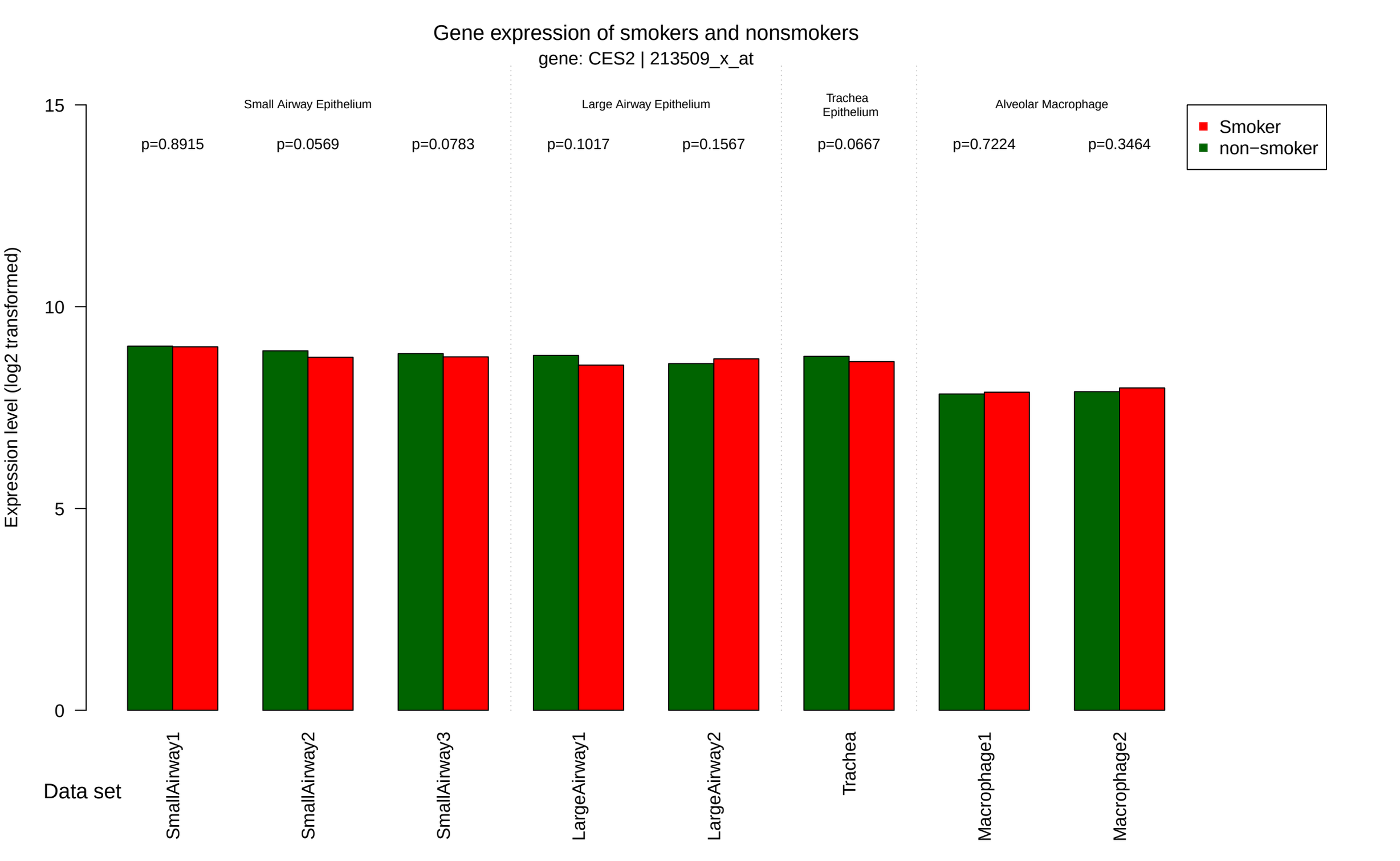
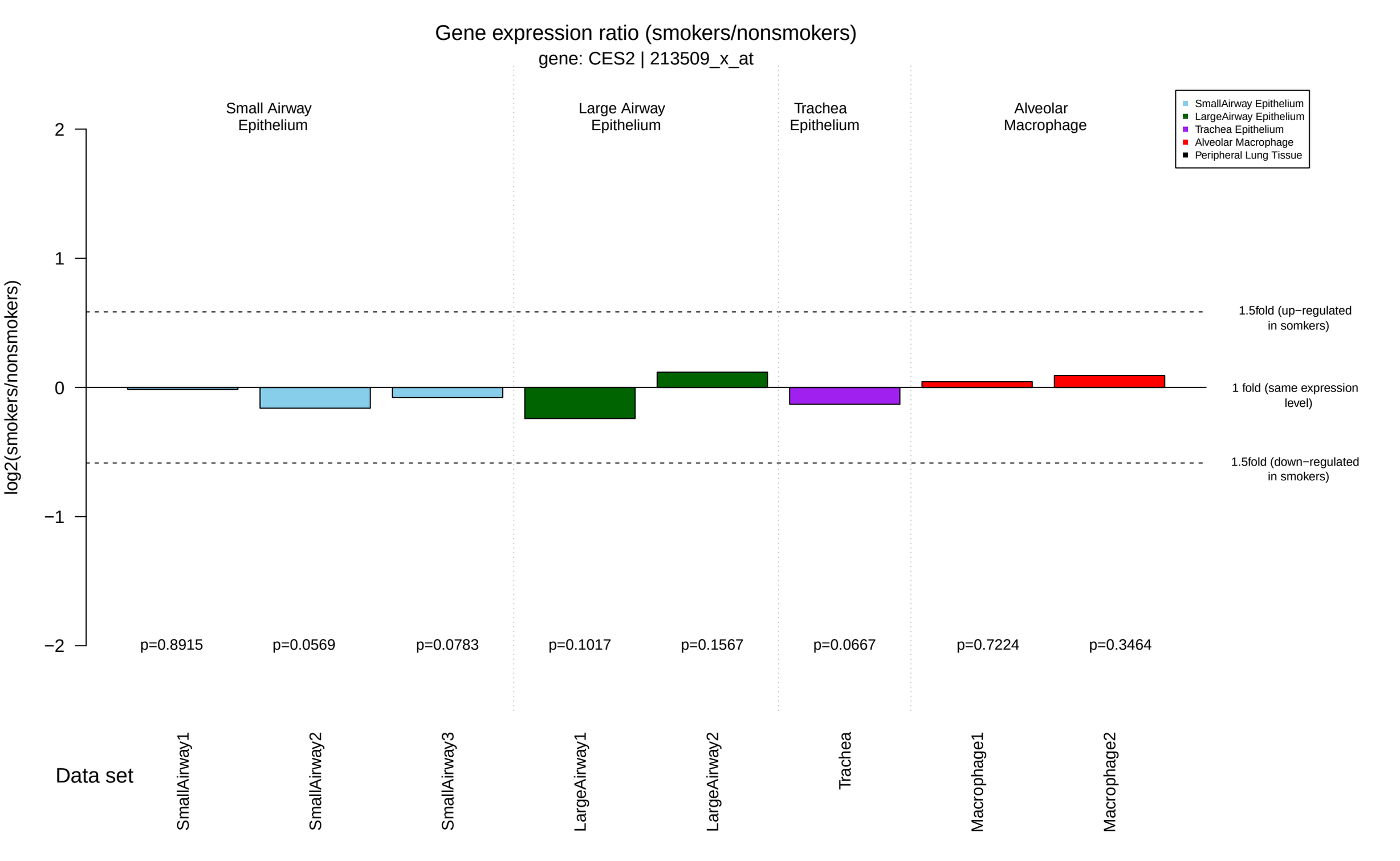
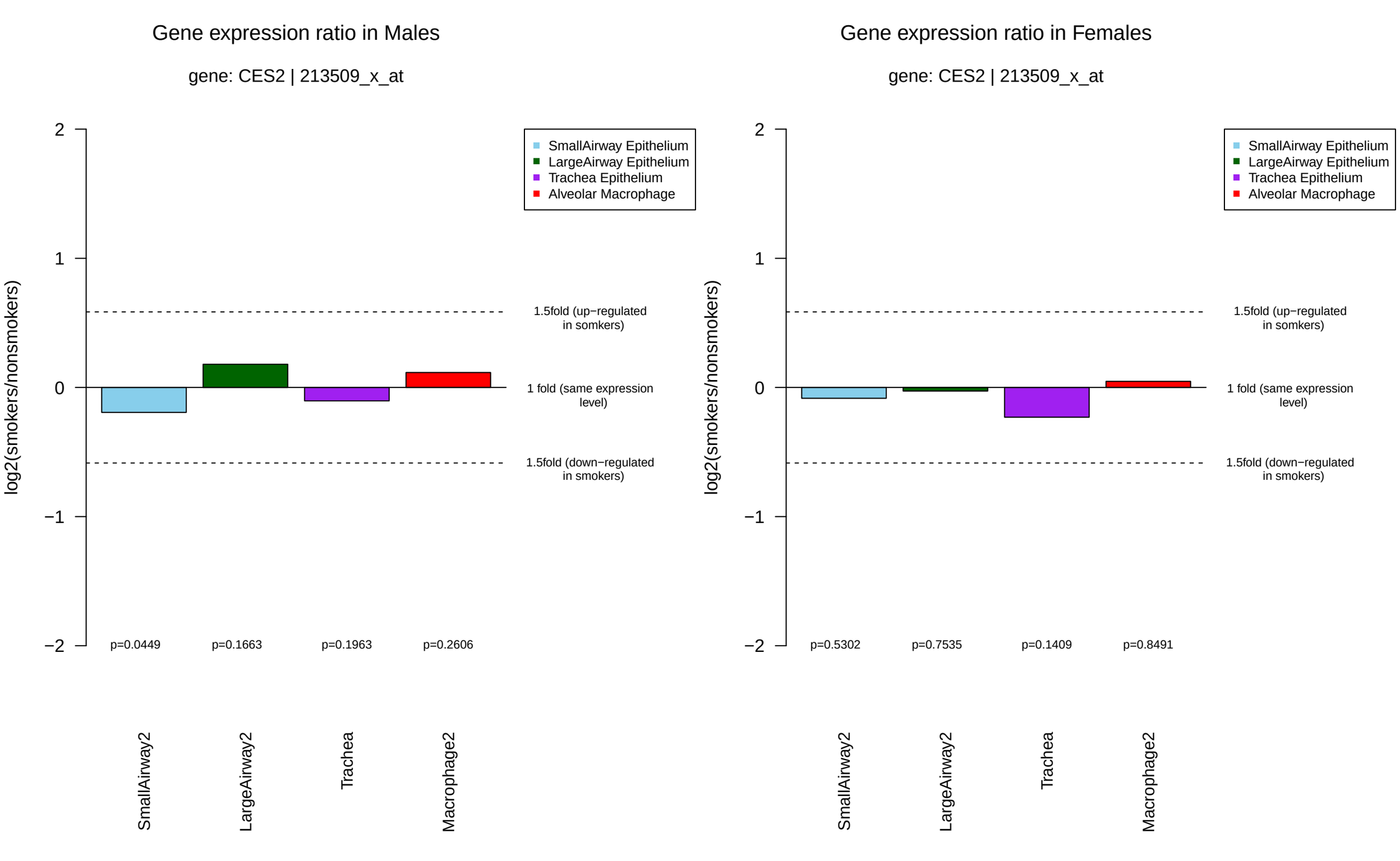
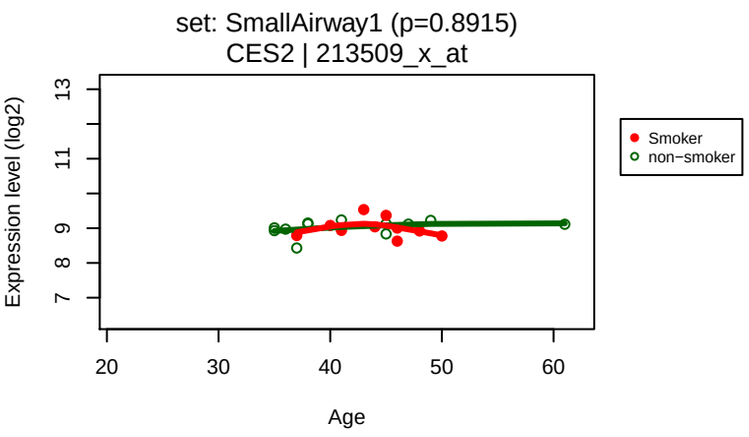
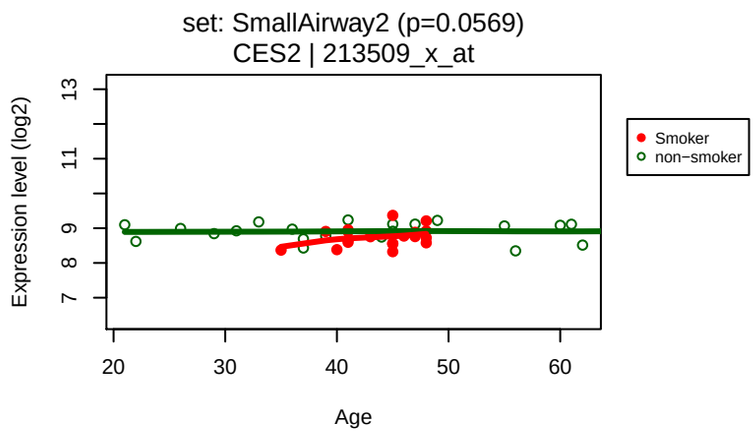
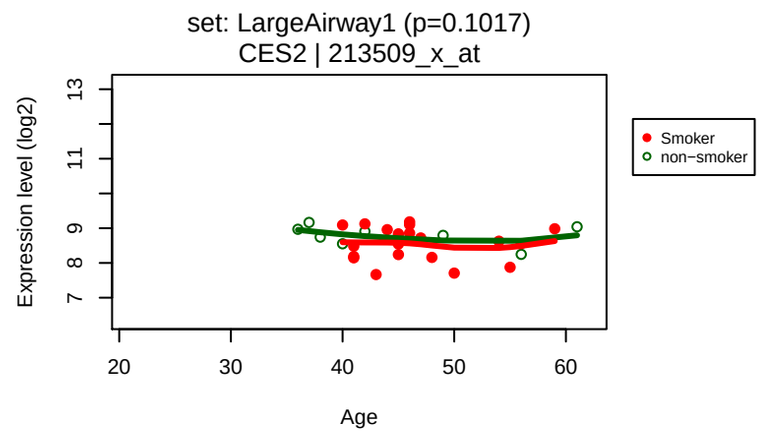

Gene expression of smokers and nonsmokers
gene: CES2 | 213509_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8915
p=0.0569
p=0.0783
p=0.1017
p=0.1567
p=0.0667
p=0.7224
p=0.3464
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

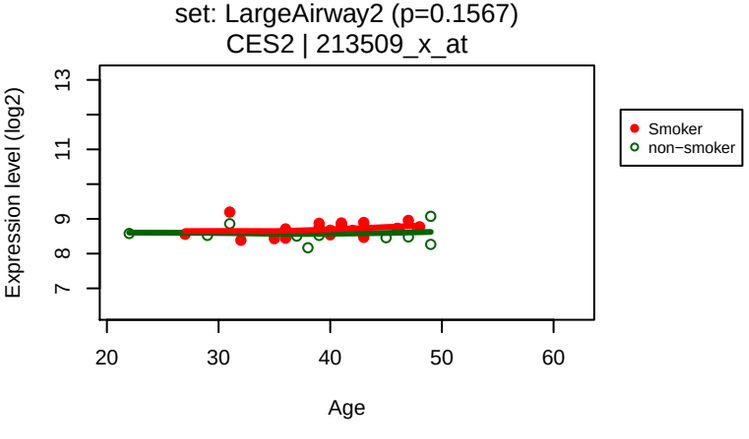

Gene expression ratio (smokers/nonsmokers)
gene: CES2 | 213509_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8915
p=0.0569
p=0.0783
p=0.1017
p=0.1567
p=0.0667
p=0.7224
p=0.3464
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

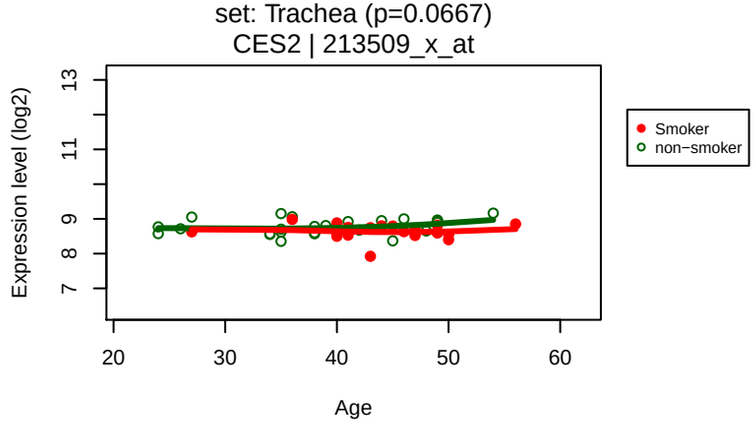

Gene expression ratio in Males
gene: CES2 | 213509_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0449
p=0.1663
p=0.1963
p=0.2606
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CES2 | 213509_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5302
p=0.7535
p=0.1409
p=0.8491
SmallAirway2
LargeAirway2
Trachea
Macrophage2

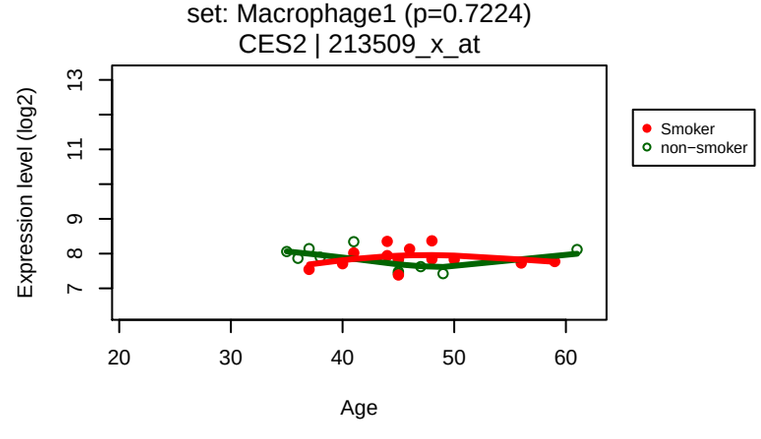

set: SmallAirway1 (p=0.8915)
CES2 | 213509_x_at
Expression level (log2)
Age
Smoker
non−smoker

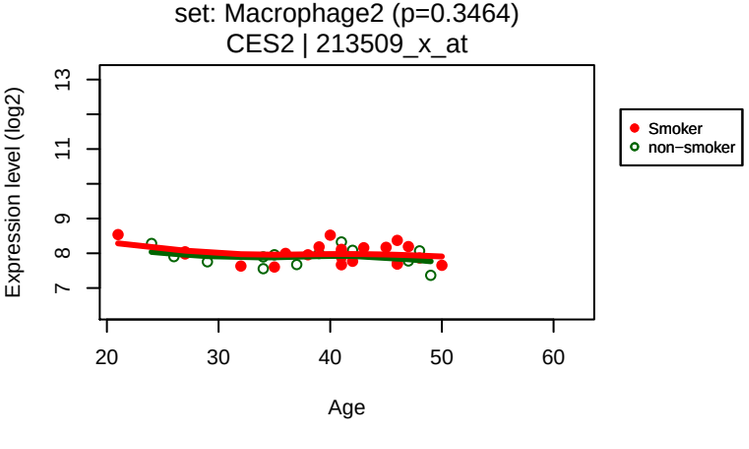

set: SmallAirway2 (p=0.0569)
CES2 | 213509_x_at
Expression level (log2)
Age
Smoker
non−smoker

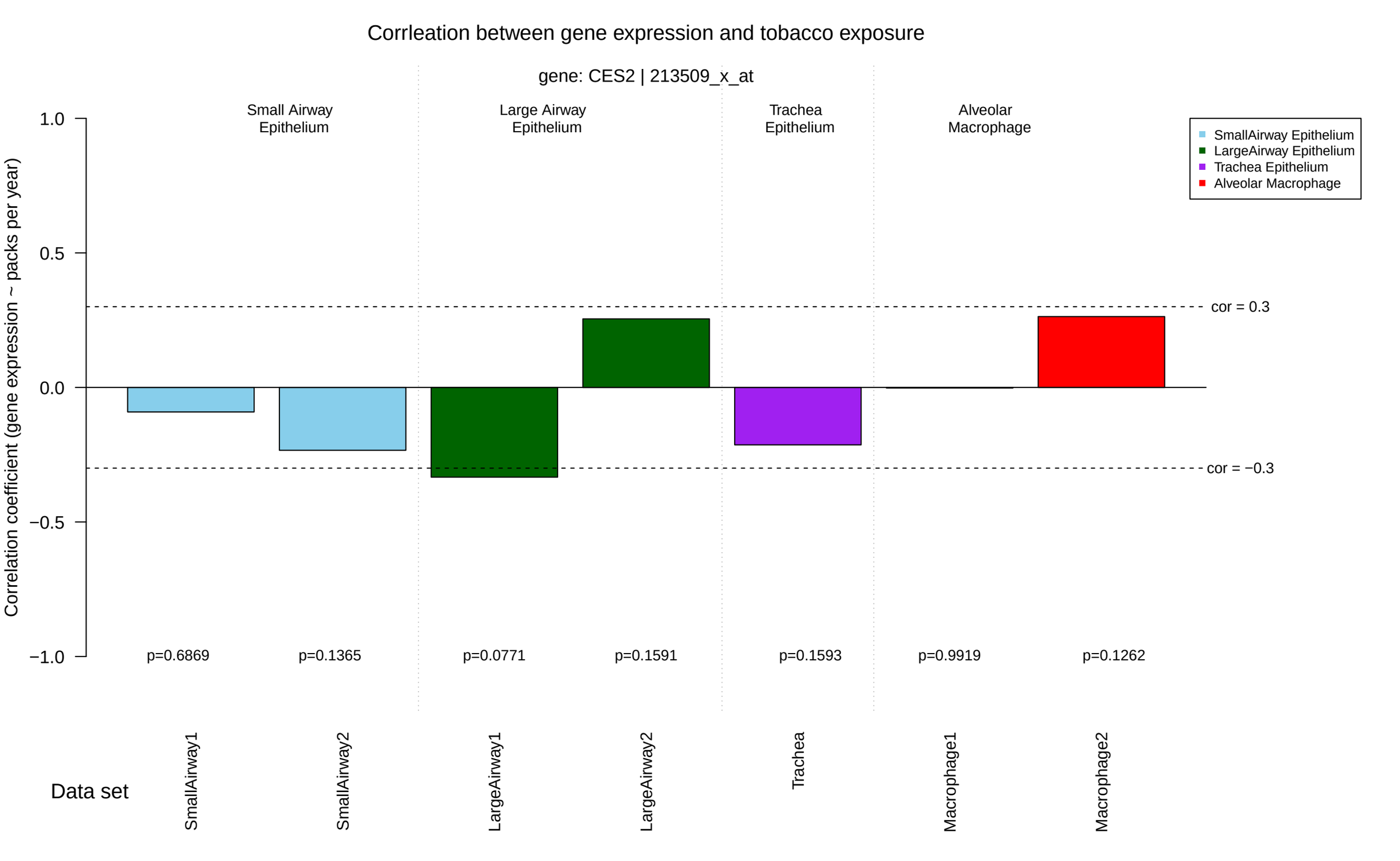

set: LargeAirway1 (p=0.1017)
CES2 | 213509_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1567)
CES2 | 213509_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0667)
CES2 | 213509_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7224)
CES2 | 213509_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3464)
CES2 | 213509_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CES2 | 213509_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6869
p=0.1365
p=0.0771
p=0.1591
p=0.1593
p=0.9919
p=0.1262
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2